MANNITOL

Inspect for crystals — use filter

Not for heart failure

CHEMICAL NAME (2R,3R,4R,5R)-hexane-1,2,3,4,5,6-hexol (D-mannitol)
CLASS Osmotic diuretic
TRADE NAMES Osmitrol · Mannitol 10% / 20% infusion · Bronchitol (inhaled)

THERAPEUTIC USES / INDICATIONS

- Raised intracranial pressure (cerebral oedema, head injury).
- Acute raised intraocular pressure (glaucoma, before eye surgery).
- Promoting diuresis in some poisonings / rhabdomyolysis (selected).
- Cystic fibrosis (inhaled).

ADMINISTRATION

- **IV infusion** via in-line filter (≤15 micron); 20% solution may crystallise — warm to dissolve, do not use if crystals remain.
- Large vein (hyperosmolar — phlebitis, extravasation necrosis).
- Urinary catheter for fluid balance.

DOSAGES

- **Raised ICP or IOP (label dose):** 1.5–2 g/kg IV over 30–60 min; smaller repeated doses are used per neuro-ICU protocol.
- **Children:** 0.5–1.5 g/kg over 30–60 min, may repeat once or twice after 4–8 h.
- Test dose (0.2 g/kg over 3–5 min) in oliguria.
- Keep serum osmolality <320 mOsm/kg.

PHARMACOKINETICS

- **Onset:** ICP falls in 15–30 min; diuresis in 1–3 h. **Duration:** 4–6 h.
- Stays in extracellular fluid; not metabolised.
- Filtered freely, not reabsorbed → excreted in urine. **t½:** 100 min.

PHARMACODYNAMICS

- ↑ plasma osmolality → draws water from brain and eye into vessels.
- Filtered by glomerulus, not reabsorbed → osmotic diuresis (water > sodium).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, nausea, chills.
- Thirst, polyuria.
- Thrombophlebitis.

ADVERSE EFFECTS (SERIOUS)

- Fluid overload / pulmonary oedema (initial volume expansion).
- Electrolyte imbalance (hypo-/hypernatraemia, hyperkalaemia), dehydration.
- Acute kidney injury (high doses, osmolality >320).
- Rebound raised ICP.

CONTRAINDICATIONS & CAUTIONS

- Anuria; severe renal failure.
- Congestive heart failure, pulmonary oedema.
- Severe dehydration.
- Active intracranial bleeding (except during craniotomy).

KEY INTERACTIONS

- **Lithium:** ↑ excretion.
- **Nephrotoxic drugs:** ↑ AKI.
- **Digoxin:** hypokalaemia/hyperkalaemia → arrhythmias.

NURSING MANAGEMENT

- Check solution for crystals; use filter.
- Monitor ICP/neuro observations (GCS, pupils).
- Strict fluid balance; hourly urine output.
- Monitor serum osmolality, Na^+, K^+, creatinine.
- Monitor lung sounds and SpO_2 (fluid overload); check IV site.

PATIENT EDUCATION

- Explain the need for frequent urine measurement.
- Report breathlessness, chest pain or pain at the drip site.

KEY POINTS

- Osmotic diuretic for raised ICP/IOP.
- Filter needle/line; check crystals.
- Label dose 1.5–2 g/kg over 30–60 min.
- Keep osmolality <320; avoid in HF.

SOURCES CHECKED **[339]** Mannitol 10% Solution for Infusion BP (UK SmPC) **[340]** Mannitol, IV (US label) **[341]** Mannitol, IV (US label, Fresenius Kabi USA)

Full references p. 432–457